

NrgOpt Photovoltaic Project Site Survey Checklist

Project Information

Item	Detail
Project / Owner Name	
Project Address	
GPS Coordinates (lat/lng)	
Owner Contact Person	
Owner Contact Phone	
Survey Date	
Survey Engineer	
Weather During Survey	

1. Building Specifications

1.1 Basic Dimensions

- ☐ Building area (m²): _
- ☐ Elevation (m): _
- ☐ Number of floors: _
- ☐ Building orientation: _
- ☐ Year built / time in use: _
- ☐ Site usage / ownership type: _

1.2 Structural Load

- ☐ Roof design live load (kg/m²): _
- ☐ Remaining capacity (kg/m²): _
- ☐ Structural reinforcement required? ☐ Yes ☐ No
- ☐ Building seismic rating: _
- ☐ Structural report available? ☐ Yes ☐ No

1.3 Roof Type

Concrete Roof: LxWxH, parapet, openings, pitch, leakage, repairs, skylights, obstacles, waterproofing

Metal Sheet Roof: profile, pitch, beam/purlin/column specs, rust, panel thickness, drawing accuracy

1.4 Obstruction Survey

Obstructions (LxWxH, direction), additional notes

1.5 Shading Analysis

Surrounding buildings, trees, chimneys, adjacent structures, winter solstice check, drone simulation

1.6 Solar Panel Layout

Rail spacing, tilt angle, row spacing, orientation, panel count, station height

2. Location & Solar Data

Parameter	Value	Method
Coordinates		[] Address lookup [] GPS
Annual Irradiance (kWh/m ²)		
Equivalent Hours (h)		
Optimal Tilt Angle (°)		
Suggested Efficiency (%)		

3. Meteorological & Environmental

3.1 Climate Data

Annual irradiation, max wind speed, snow load, extreme temps, rainfall, lightning frequency

3.2 Environmental Risks

Coastal proximity, nearby industrial plants, flood risk, seismic zone, typhoon path, soil corrosion, bird activity

4. Electrical Infrastructure

4.1 Grid Connection

Connection type, transformer count/capacity/model, protection devices, existing PV, distance to connection point

4.2 Distribution Room

Location, cabinet space, spare breakers, cable trench, road crossing, single-line diagram

4.3 Consumption Profile

Generation mode, contracted demand, hourly load profile, peak period, weekend/night operation

4.4 Monthly Consumption (kWh)

Month	kWh	Month	kWh
Jan		Jul	
Feb		Aug	
Mar		Sep	
Apr		Oct	
May		Nov	
Jun		Dec	

Annual total: _ kWh

4.5 Tariff Information

Supply company, tariff category, weighted average rate, demand charge

4.6 Time-of-Use Pricing

Period	Rate (CNY/kWh)	Start	End
Critical Peak			
Peak			
Shoulder			
Off-Peak			

5. Site Access & Logistics

5.1 Access Assessment

Road width, truck access, crane radius/parking, road closure, material staging, lifting method, construction power/water

5.2 Safety Considerations

Roof edge protection, access point, fall protection anchors, evacuation route, hazardous materials

6. Regulatory & Documentation

6.1 Owner Documentation

Property ownership, land use, construction drawings, structural calculation, electrical diagram, fire safety, EIA, business license, CAD/PDF files

6.2 Regulatory Check

Local PV filing, grid pre-approval, height restriction, heritage area, aviation restriction, neighbour consent

7. Photo Checklist (20 items)

N/S/E/W views, roof panorama, close-ups, obstructions, parapet, equipment, distribution room, transformer, cable route, entrance, drone video, GPS photo, team photo

8. Cost & Timeline

Cable length, trenching, special equipment, grid outage window, construction start, duration, seasonal constraints

9. Sign-off

Role	Name	Signature	Date
Survey Engineer			
Project Manager			
Owner Representative			
Electrical Engineer			
Structural Engineer			

10. Additional Notes
